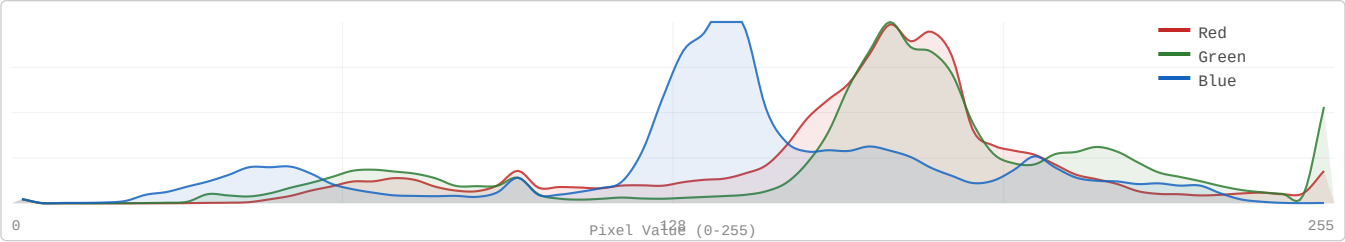
Decorrelation Gap: KODIM12

768x512 | 24-bit RGB | PNG (lossless)

Farm Fence and Trees | Strongly one-dimensional (PC1 = 95.09%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9133	0.8721
G	0.9133	1.0000	0.9674
B	0.8721	0.9674	1.0000

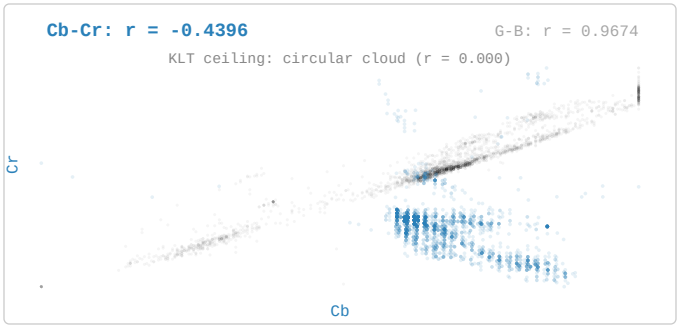
Cb-Cr: -0.440

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.1005	-0.5092
Cb	-0.1005	1.0000	-0.4396
Cr	-0.5092	-0.4396	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.913293	-0.100517	0.1005
R-B → Y-Cr	0.872110	-0.509158	0.5092
G-B → Cb-Cr	0.967415	-0.439553	0.4396
Avg |r|	0.917606	0.349742	0.3497

Decorrelation Gap: KODIM12

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Farm Fence and Trees | Strongly one-dimensional (PC1 = 95.09%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

61.9% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

4.2%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	45.42 dB
G	49.26 dB
B	44.97 dB
Overall	46.18 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.4937	-0.6434	-0.5850	5966.22
PC2	-0.8389	0.1752	0.5153	245.82
PC3	0.2290	-0.7452	0.6263	62.55

8. PCA Derived Metrics

Condition Number

95.39

PC1/PC2: 24.27 | PC2/PC3: 3.93

Blue Channel Independence

4.2%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.64)

9. Redundancy Profile

Classification: Strongly one-dimensional | CN: 95.39 | PC1: 95.1%

RGB Correlation: Avg |r| = 0.917606 | R-G: 0.9133 | R-B: 0.8721 | G-B: 0.9674

YCbCr Residual: Avg |r| = 0.349742 | Y-Cb: -0.1005 | Y-Cr: -0.5092 | Cb-Cr: -0.4396

Efficiency: 61.9% of KLT ceiling | Gap: 0.3497 avg |r| remaining

Blue Channel: 4.2% independent | 4:2:0 PSNR: 46.18 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.